CARLO YEHUDA MELONI

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RESEARCH INTERESTS

Language evolution; linguistic typology; historical and Semitic linguistics; Bayesian phylogenetics; Gaussian processes and spatial/areal modeling; computational and quantitative linguistics; clause-linkage syntax.

EMPLOYMENT HISTORY

Postdoctoral Researcher 2025 – present (contract through 2027)
Chair of Multilingual Computational Linguistics, University of Passau. Contributed to collaborative research on cross-linguistic naming variation, Semitic root phonotactics, and phonetic annotation software. Converted linguistic datasets to Cross-Linguistic Data Formats and provided statistical modeling and analysis support for departmental research projects.

Research Associate 2021 – 2025
Distributional Linguistics Lab, Institute for the Interdisciplinary Study of Language Evolution, University of Zurich. Salaried doctoral researcher in the SYNPLEX project (*The Evolution of Syntactic Complexity*), under Prof. Balthasar Bickel. Lead manuscript drafter and principal analyst for three collaborative studies on syntactic hierarchy, writing, and the syntax–semantics interface. Implemented Bayesian phylogenetic analyses in R/Stan, including Gaussian-process and continuous-time Markov models.

EDUCATION

Ph.D. in Linguistics 2021 – 2025
University of Zurich, Switzerland
Advisors: Prof. Balthasar Bickel, PD Dr. Chundra A. Cathcart, Prof. Jessica K. Ivani
"Syntactic Hierarchy in Linguistic Evolution: A Phylogenetic View of Clause-Linkage"
Final grade: 6.0/6.0 (summa cum laude)

M.A. in Linguistics 2019 – 2021
Tel Aviv University, Israel
Advisors: Dr. Evan Gary Cohen, Prof. Hezy Mutzafi
"The Reš Riddle: Identifying the Biblical Hebrew Rhotic"
Final grade: 94/100 (magna cum laude)

B.A. in Linguistics (minor in Biology) 2016 – 2019
Tel Aviv University, Israel
Final grade: 96/100 (summa cum laude)

PUBLICATIONS

Published articles and conference proceedings

Chundra Cathcart, Arne Rubehn, Katja Bocklage, Luca Ciucci, Kellen Parker van Dam, Alžběta Kučerová, Jekaterina Mažara, **Meloni, Carlo Y.**, David Snee, and Johann-mattis List (2026). "How communicatively optimal are exact numeral systems? Once more on lexicon size and morphosyntactic complexity". In: *Proceedings of the Annual Meeting of the Cognitive Science Society*. Vol. 48. URL: https://escholarship.org/uc/item/4hz9j5f8.

Kellen Parker van Dam, David Snee, Johann-Mattis List, Andre Goderich, Luca Ciucci, **Meloni, Carlo Y.**, Keen Thaam, Nora Josefina Muheim, Katja Bocklage, and Arne Rubehn (2026). "Besra: An Interactive Tool for Speech Annotation, Phonetic Analysis, and Computer-Assisted Fieldwork". In: *Open Research Europe* 6, 420. Version 1; awaiting peer review. doi:10.12688/openreseurope.25173.1.

Meloni, Carlo Y. and Johann-Mattis List (Dec. 2025). "Towards a Unified Conversion Table for Semitic Transcriptions and Transliterations". In: *Computer-Assisted Language Comparison in Practice: Tutorials on Computational Approaches to the History and Diversity of Languages* 8.2, pp. 107–112. doi:10.15475/cal-cip.2025.2.6.

Meloni, Carlo Y., Shauli Ravfogel, and Yoav Goldberg (June 2021). "Ab Antiquo: Neural Proto-language Reconstruction". In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, pp. 4460–4473. doi:10.18653/v1/2021.naacl-main.353.

Accepted for publication

Meloni, Carlo Y. and Kellen Parker van Dam (2026). "A New Look at Avoidance Phenomena in Maltese Roots". In: *Journal of Maltese Studies*. Accepted manuscript available on Zenodo. doi:10.5281/zenodo.23450593. Forthcoming.

Meloni, Carlo Y., Jessica K. Ivani, Taras Zakharko, Guanghao You, Kellen Parker van Dam, Chundra A. Cathcart, and Balthasar Bickel (2026). "No Evidence for a Universal Effect of Writing on Language Use or the Evolution of Syntax". In: *Proceedings of the National Academy of Sciences*. Forthcoming.

Manuscripts under peer review (preprints available)

Alžběta Kučerová, **Meloni, Carlo Y.**, and Johann-Mattis List (Aug. 2026). "Naming variation across languages is shaped by shared histories, semantics and familiarity". In: *PsyArXiv*. doi:10.31234/osf.io/8tq2p_v1.

Meloni, Carlo Y., Chundra Cathcart, Erik J. Ringen, Jessica K. Ivani, and Balthasar Bickel (July 2025). "A Phylogenetic Perspective on the Syntax-Semantics Interface: A Case Study in Clause Linkage Constructions". In: *bioRxiv*. doi:10.1101/2025.07.19.665697.

Meloni, Carlo Y., Chundra A. Cathcart, Jessica K. Ivani, Taras Zakharko, Erik J. Ringen, and Balthasar Bickel (July 2025). "Linguistic evolution minimizes the amount of syntactic hierarchy in grammar". In: *bioRxiv*. doi:10.1101/2025.07.19.665693.

SELECTED INVITED TALKS & CONFERENCE PRESENTATIONS

A Quantitative Analysis of Biblical Hebrew's Rhotic 2026
Invited lecture, European Hebrew Journal International Scholarly Conference (scheduled for November 2026).

A Phylogenetic Perspective on the Syntax–Semantics Interface 2026
Paper, 16th International Conference of the Association for Linguistic Typology.

Word Family Evolution in Semitic Languages 2026
Workshop talk, *Quantitative Cross-Linguistic Explorations of Space, Time, and Evolution* (qc-leste).

Writing Does Not Impact the Evolutionary Dynamics of Syntax 2025
Paper, Emerging Topics in Typology, Max Planck Institute for Evolutionary Anthropology; poster, Societas Linguistica Europaea.

The Resh Riddle: Identifying the Biblical Hebrew Rhotic 2025
Poster, 8th 'R-atics Conference.

Evolutionary Dynamics of Clause Linkage Constructions 2024
Invited colloquium, University of Passau.

Ab Antiquo: Neural Proto-language Reconstruction 2021
Paper, NAACL.

AWARDS & GRANTS

Yohanan Goldman Scholarship for Outstanding M.A. Students 2020
Tel Aviv University, Lester and Sally Entin Faculty of Humanities.

Travel Grant 2019

Department of Linguistics, Tel Aviv University.

Scholarships for Academic Excellence; Dean's List 2017, 2018
Tel Aviv University, Lester and Sally Entin Faculty of Humanities.

Raffaele Cantoni Fellowship for Undergraduate Students 2016
Hebrew University of Jerusalem.

TEACHING (ASSISTANT ROLE)

Fieldwork Methods and Elicitation Techniques (M.A.), University of Zurich Spring 2024
Methods in Evolutionary Linguistics (M.A.), University of Zurich Fall 2022, 2023
Advanced Phonology (B.A.), Tel Aviv University Spring 2020
Introduction to Linguistics (B.A.), Tel Aviv University Fall 2019

ADDITIONAL RESEARCH EXPERIENCE

Morphology and Syntax Expert 2021
Israel Innovation Authority: morphological segmentation, POS tagging, and dependency annotation of a Hebrew corpus using Universal Dependencies v2.8.

Mandaic Transcription Expert 2017 – 2021
Tel Aviv University: transcription and annotation of Mandaic texts for the Comprehensive Aramaic Lexicon project (PI: Prof. Matthew Morgenstern).

Research Assistant 2020
Tel Aviv University: diachronic corpus searches and data coding for Prof. Mira Ariel.

TECHNICAL SKILLS

Programming:	R, Python, SQL
Statistical modeling:	Stan, Bayesian hierarchical modeling, Gaussian processes
Phylogenetics:	BEAST2, Bayesian phylogenetic comparative methods
Research computing:	Git, Linux, Slurm/HPC
Reproducibility:	Quarto, R Markdown, Pandoc, LaTeX

ACADEMIC SERVICE

Peer reviewer for *Linguistic Typology*, *Scientific Reports*, and *Journal of Under-Resourced Languages*.

SELECTED ADVANCED TRAINING

Gaussian Process and Uncertainty Quantification Summer School (Manchester, 2026); Bayesian Phylogenetic Inference with BEAST2 (2024); Statistical Methods for Linguistics and Psychology (Potsdam, 2023 and 2024); Lisbon Machine Learning School (2022).

LANGUAGES

Modern: Italian and Modern Hebrew (native); English (fluent); German, French, Spanish, Esperanto (proficient); Yiddish, Russian (conversational); Modern Standard Arabic, Palestinian Arabic (passive).
Historical and textual (reading): Latin; Biblical and Mishnaic Hebrew; Biblical and Targumic Aramaic; Jewish Babylonian Aramaic; Syriac; Mandaic; Old Babylonian; Ge'ez.